

SQL Mastery Roadmap (10 Days)

Day 1 - SQL Fundamentals

Topics:

1. What is Database?
 2. What is RDBMS?
 3. SQL vs NoSQL
 4. Tables, Rows, Columns
 5. PostgreSQL Installation
 6. SQL Syntax
 7. Data Types
 8. Basic Operators
 9. SELECT Statement
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Day 2 - Filtering Data

Topics:

- WHERE
 - AND
 - OR
 - NOT
 - IN
 - BETWEEN
 - LIKE
 - LIMIT
 - DISTINCT
 - ORDER BY
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Day 3 - DDL Commands

Topics:

- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- TRUNCATE TABLE
- Constraints

- PRIMARY KEY
 - FOREIGN KEY
 - UNIQUE
 - NOT NULL
 - CHECK
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Day 4 - DML Commands

Topics:

- INSERT
- UPDATE
- DELETE

Mini Project:

Employee Management Database

Day 5 - Aggregate Functions

Topics:

- COUNT()
 - SUM()
 - AVG()
 - MIN()
 - MAX()
 - GROUP BY
 - HAVING
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Day 6 - JOINS (Most Important)

Topics:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- SELF JOIN
- CROSS JOIN

Day 7 - Subqueries

Topics:

- Scalar
 - Row
 - Column
 - Table Subqueries
 - Nested Queries
 - Correlated Queries
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Day 8 - Functions

Topics:

String Functions

- CONCAT
- LENGTH
- SUBSTRING
- REPLACE
- UPPER
- LOWER

Numeric Functions

- ABS
- MOD
- FLOOR
- ROUND
- CEILING

Date Functions

- DATE
 - TIME
 - TIMESTAMP
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Day 9 - Advanced SQL

Topics:

- Views
 - Indexes
 - Transactions
 - ACID
 - COMMIT
 - ROLLBACK
 - SAVEPOINT
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Day 10 - Expert SQL

Topics:

- Window Functions
 - ROW_NUMBER
 - RANK
 - DENSE_RANK
 - LEAD
 - LAG
 - CTE
 - Recursive Queries
 - Query Optimization
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SQL Portfolio Projects

Project 1

Student Management System

Tables:

- Students
- Courses
- Enrollments
- Faculty

Concepts:

- CRUD

- Joins
 - Aggregations
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Project 2

Employee Management System

Tables:

- Employees
- Departments
- Salary

Concepts:

- Constraints
 - Joins
 - Group By
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Project 3

E-Commerce Database

Tables:

- Users
- Products
- Orders
- Payments

Concepts:

- Advanced Joins
 - Transactions
 - Views
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Project 4

Hospital Management System

Tables:

- Patients
- Doctors
- Appointments
- Medicines

Concepts:

- Foreign Keys
- Complex Queries